NICHOLAS A **LOMBARDO**, PHD

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Education

2025	Ph.D.	Earth and Planetary Sciences	Yale University
	Dissertation: "The Dynamics and Composition of the Middle Atmosphere of Titan"		
2021	M.Phil.	Earth and Planetary Sciences	Yale University
2017	B.S.	Physics & Mathematics	Central Conn. State Univ.

Additional Training

2024	Mission Design School: Planetary Science (PI)	10 wk	Caltech Jet Propulsion Lab
2024	Tracer Mixing in Fluids Summer School	2 wk	UMD Brin Center of Mathematics

Professional Appointments

2023–present	Visiting Lecturer	Trinity College Dept. of Physics (2 semesters)
	"	Central CT State Univ. Dept. of Physics (5 semesters)
	"	CT State Tunxis Dept. of Science (1 semester)
	"	University of Hartford Dept. of Physics (1 semester)
2026–2026	Post Doctoral Researcher	NASA Goddard Space Flight Center through SURA
2019–2025	Research & Teaching Fellow	Yale University Dept. of Earth & Planetary Sciences
2017–2019	Research Assistant	NASA Goddard Space Flight Center through UMBC
2016–2017	Intern	NASA Goddard Space Flight Center
2013–2016	Secondary-school STEM Tutor	Central Connecticut State Univ. TRiO Office

Space Mission Involvement

2025–present	Titan Science Case Dev. Team	*Habitable Worlds Observatory*
2019–2025	Student Associate	*Dragonfly* Mission to Titan (NASA New Frontiers 4)
2016–2019	Research Assistant	*Cassini* Composite Infrared Spectrometer Team

Honors and Awards (underline – financial awards)

2024 Yale Estwing Hammer Prize – *Outstanding Graduate Student in Earth and Planetary Sciences*
2019 Yale Geology and Geophysics Bateman Fellowship
2018 NASA Group Achievement Award – *Cassini CIRS Investigation Team for Grand Finale Orbits*
2018 NASA Group Achievement Award – *Cassini CIRS Team for Solstice, F-Ring, and Proximal Orbits*
2017 CCSU Physics and Engineering Physics Departmental Honors
2016 CCSU John Bulman Award – *Service to the Department of Physics and Engineering Physics*
2013 CCSU-NSF Computer Science, Math, and Physics Scholarship (4 years)

University Teaching

Instructor of Record

Fall 2026	17 students	Trinity College	Phys 101 Principles of Physics
"	30 students	"	Phys 101 Principles of Physics Laboratory
"	9 students	CT State Tunxis.	Phys 1100 Introductory Physics Laboratory
"	30 students	Central CT State Univ.	Phy 121 General Physics I Laboratory
"	25 students	"	Phy 125 Univ. Physics I Laboratory
"	30 students	"	Phy 111 Intro. To Physics Laboratory
Spring 2026	17 students	Trinity College	Phys 231 Electricity and Magnetism Laboratory
"	30 students	University of Hartford	Phy 102 Electricity and the Human Body
"	25 students	"	Phy 102 Electricity and the Human Body
"	30 students	"	Phy 122 Physics for Architecture

"	17 students	Central CT State Univ.	Phys 121 General Physics I Laboratory
"	34 students	"	Phys 121 General Physics I
"	17 students	"	Phys 126 University Physics II Laboratory
Fall 2024	17 students	"	Phys 126 University Physics II
Spring 2024	16 students	"	Phys 126 University Physics II Laboratory
Spring 2023	17 students	"	Phys 122 General Physics II Laboratory

Graduate Teaching Fellow

Spring 2025	13 students	Yale Univ.	ENV 774 Framing Climate Migration Science/Policy/Ethics
Fall 2023	120 students	"	EPS 274 Fossil Fuels and World Energy
Fall 2019	27 students	"	G&G 140 Intro. to Atmosphere, Ocean, and Climate Change

Competed Grants

Research Grants

2024 NSF Astro.: *Investigating the Chemical and Dynamical Complexity of Titan's Upper Atmosphere* (Collaborator)

2024 NASA Solar System Observations: *Winds of change: a multi-decade study of Titan's middle atmosphere across seasons* (Collaborator)

Awarded Astronomical Observing Proposals

2024 7 hr Atacama Large Mm/sub-mm Array: *Winds of Change: Temporal Variability of Titan's High-Altitude Circulation System* (Co-I; PI: Cordiner)

2023 12 hr Hale Telescope, Palomar Observatory: *Characterizing Titan's stratospheric wind profile via adaptive optics imaging of a stellar occultation* (Collaborator; PI: Marlin)

2022 3 hr Faulkes Telescope N., Las Cumbres Observatory: *Constraining Titan's Stratospheric Winds near Equinox via a Stellar Occultation* (Principal Investigator)

2022 12 hr NASA Infrared Telescope Facility: *Investigating Nitrile Chemistry on Titan* (Co-I; PI: Coustenis)

Travel Grants

2025 AAS Division for Planetary Sciences Hartmann Student Travel Grant

2025 Caltech Earth and Planetary Clouds Workshop Early-Career Travel Grant

2022 Europlanet Science Congress Student Travel Grant

2021 Titan Through Time V Student Travel Grant

2019 AAS Division for Planetary Sciences Hartmann Student Travel Grant

2016 CCSU Faculty Student Research Grant & CCSU Alumni Travel Grant

Refereed Publications

2026. Wright, L., Teanby, N. A., Irwin, P. G. J., Nixon, C. A., **Lombardo, N. A.**, Lora, J. M., Ford, J. S., "Titan's N-S Asymmetry Boundary explained using High-Resolution Mapping from Cassini/CIRS", under review in *Nature Communications*

2026. Cordiner, M. A., Lian, Y., Lora, J. M., **Lombardo, N. A.**, Wright, L., Nosowitz, J., Cosentino, R., Teanby, N. A., Nixon, C. A., Thelen, A. E., Newman, C. E., Charnley, S. B., Young, E. F., "Temporal Variability of Titan's Stratospheric Winds from Southern Fall to Late Winter (2016 – 2023), *The Planetary Science Journal*, **XX**, XX

2026. **Lombardo, N. A.** & Lora, J. M., "Interaction of Trace Species and Three-Dimensional Circulation in Titan's Middle Atmosphere", *Journal of Geophysical Research: Planets*, **131**, 9

2026. Sinclair, J. A., Greathouse, T. K., Giles, R. S., Sung, K., Nixon, C. A., **Lombardo, N. A.**, Hue, V., Moses, J. I., Fletcher, L. N., Irwin, P. G. J., Orton, G. S., "Detection of propadiene (CH_2CCH_2), propene (C_3H_6), and non-detection of propane (C_3H_8) in Jupiter's Northern Polar Stratosphere", *Icarus*, **117156**

2026. Shultis, J., Waugh, D. W., Toigo, A. D., **Lombardo, N. A.**, Lora, J. M., "Impact of Titan's Polar Vortex on the Transport of Chemical Species", *Journal of Geophysical Research: Planets*, **131**, 2

2025. Marlin, T. C., Young, E. F., de Kleer, K., Cordiner, M., **Lombardo, N. A.**, de Pater, I., Lora, J. M., Corlies, P., Cosentino, R., Nixon, C. A., Rodgriguez, S., Thelen, A., "Zonal Winds in Titan's Middle Atmosphere from a Stellar Occultation Observed with Keck Adaptive Optics", *The Planetary Science Journal*, **6**, 286

2025. Wright, L., Teanby, N. A., Irwin, P. G. J., Nixon, C. A., **Lombardo, N. A.**, Mitchell, D., Lora, J. M. "Seasonal Evolution of Titan's Stratospheric Tilt and High-Resolution Temperatures, Zonal Winds, and Polar Vortices", *The Planetary Science Journal*, **6**, 114

2025. Nixon, C. A., Bézard, B.†, Cornet, T.†, Coy, B. P.†, de Pater, I.†, Es-Sayeh, M.†, Hammel, H. B.†, Lellouch, E.†, **Lombardo, N. A.†**, López-Puertas, M.†, Lora, J. M.†, Rannou, P.†, Rodriguez, S.†, Teanby, N. A.†, Turtle, E. P.†, +30 others, "Titan's Atmosphere in Late Northern Summer from JWST and Keck Observations", *Nature Astronomy, †These authors contributed equally*

2024. Bézard, B., Nixon, C. A., Vinatier, S., Lellouch, E., Greathouse, T. K., Jiles, R., **Lombardo, N. A.,** Jolly, A., "The D/H ratio in Titan's acetylene from high spectral resolution IRTF/TEXES observations", *Icarus*, **421**, 116221

2024. Light, S., Gurwell, M., Thelen, A., **Lombardo, N. A.**, Nixon, C. A., "Measurements of Titan's Stratospheric Winds During Equinox with the eSMA", *The Planetary Science Journal*, **5**, 98

2023. **Lombardo, N. A.** & Lora, J. M., "The Heat and Momentum Budgets of Titan's Middle Atmosphere", *Journal of Geophysical Research: Planets*, **128**, e2023JE00806

2023. Lewis, N. T., **Lombardo, N. A.**, Read, P. T., Lora, J. M., "Equatorial Waves and Superrotation in the Stratosphere of a Titan General Circulation Model", *The Planetary Science Journal*, **4**, 149

2023. Coy, B. P., Nixon, C. A., Rowe-Gurney, N., Achterberg, R. K., **Lombardo, N. A.**, Fletcher, L. N., Irwin, P. G. J. "Spitzer IRS Observations of Titan as a Precursor to JWST MIRI Observations", *The Planetary Science Journal*, **4**, 114

2023. **Lombardo, N. A**. & Lora, J. M., "Influence of Observed Seasonally Varying Composition on Titan's Stratospheric Circulation", *Icarus*, **390**, 115291

2022. Steffans, B.L., Nixon, C.A, Sung, K., Irwin, P.G.J., **Lombardo, N.A.**, Pereira, E. "New Constraints on Titan's Stratospheric n-Butane Abundance", *The Planetary Science Journal,* **3**, 59

2020. Hewett, D., Bernath, P. F., Wong, A., Billinghurst, B. E., Zhao, J., **Lombardo, N. A.**, Nixon, C. A., Jennings, D. E. "N_2 and H_2 Broadened Isobutane Infrared Absorption Cross Sections and Butane Upper Limits on Titan", *Icarus,* **344**, 113460

2019. Nixon, C. A., Ansty, T. M., **Lombardo, N. A.**, Bjoraker, G. L., Achterberg, R. K., Annex, A. M., Rice, M., Romani, P. N., Jennings, D. E., Coustenis, A., Bézard, B., Vinatier, S., Lellouch, E., Teanby, N. A., Cottini, V., Flasar, F. M. "*Cassini* Composite Infrared Spectrometer Observations of Titan 2004-2017", *The Astrophysical Journal Supplement Series*, **244**, 14

2019. ***Lombardo, N A.**, Nixon, C. A., Greathouse, T. K., Bézard, B., Jolly, A., Vinatier, S., Teanby, N. A., Richter, M. J., Irwin, P. J. G., Coustenis, A., Flasar, F. M. "Detection of Propadiene on Titan", *The Astrophysical Journal Letters*, **881**, L33 - **featured on Sci-News and NASA Astrobiology Institute homepage*

2019. **Lombardo, N. A.**, Nixon, C. A., Sylvestre, M., Jennings, D. E., Teanby, N., Irwin, P. G. J., and Flasar, F. M. "Ethane in Titan's Stratosphere from *Cassini* CIRS Far- and Mid-Infrared Spectra", *The Astronomical Journal*, **157**, 160-170

2019. **Lombardo, N. A.**, Nixon, C. A., Achterberg, R. K., Sung, K., Jolly, A., Irwin, P. G. J., and Flasar, F. M. "Spatial and Seasonal Variations in C_3H_x Hydrocarbon Abundance in Titan's Stratosphere with *Cassini* CIRS Observations", *Icarus*, **317**, 454 – 469

Professional Service

University Service

2021–2024 Student representative on Yale Earth and Planetary Sciences Colloquium Committee

2020–2021 Undergraduate & Graduate Student Mentor, Yale Dept. of Earth and Planetary Sciences
2014–2016 President of the Central CT State Univ. Society of Physics Students

Conference Service

2025	Co-convener: Exploration of Titan;	EPSC-DPS Joint Meeting
2024	Virtual Session Chair: Io;	AAS Division for Planetary Sciences 56th Meeting
2024	Session Chair: Venus;	AAS Division for Planetary Sciences 56th Meeting
2024	Science Organizing Committee member;	AAS Division for Planetary Sciences 56th Meeting
2017	Poster Session Lead;	Titan Through Time IV Workshop

Journal Referee

The Astronomical Journal, The Astrophysical Journal, Geophysical Research Letters, New Astronomy, The Planetary Science Journal, Yale Undergraduate Research Journal

Proposal Review Panels

Panelist, secretary, and external reviewer for NASA Planetary Science

Invited Colloquia and Seminars

2026 Depts of Earth & Space Sciences, Physics & Engineering Physics Joint Seminar, Central CT State Univ.
2025 School of Earth Sciences Seminar, University of Bristol
2025 Planetary Science Colloquium, Southwest Research Institute, Boulder, CO
2025 Yuk Lunch Seminar, Department of Geology and Planetary Science, Caltech
2025 Departments of Physics & Astronomy and Earth & Climate Science Joint Seminar, Bates College
2024 Department of Earth and Planetary Sciences Seminar, Johns Hopkins University
2023 Department of Mathematics Colloquium, Central Connecticut State University
2022 Department of Geological Sciences Seminar, Central Connecticut State University
2021 Department of Physics and Engineering Physics Seminar, Central Connecticut State University
2019 School of Physics and Astronomy Space Physics Seminar, University of Leicester
2018 Center for Research in Exploration in Space Science and Tech. Seminar, NASA GSFC
2018 Department of Geological Sciences Seminar, Central Connecticut State University

First-Authored Conference and Workshop Talks

2025. **Lombardo, N. A.***, de Batz de Trenquelleon*, B., Shultis, J., Lora, J. M., Rannou, P., Waugh, D., Lian, Y., Newman, C. "The Titan Middle Atmosphere Intercomparison Project", Europlanet Science Congress – Division for Planetary Sciences Joint Meeting **--these authors contributed equally*

2025. **Lombardo, N. A.** & Lora, J. M., "Simulating Clouds in the Middle Atmosphere of Titan", Caltech Earth and Planetary Clouds Workshop

2024. **Lombardo, N. A.** & Lora, J. M. "Feedbacks between Idealized Chemical Tracers and Dynamics through Radiative Heating in Titan's Middle Atmosphere", 55th Division for Planetary Sciences Meeting

2023. **Lombardo, N. A.**, et al., "Measurement of Hydrogen Isocyanide in Titan's Lower Stratosphere", American Geophysical Union

2023. **Lombardo, N. A.** & Lora, J.M. "Simulating the Production of Stratospheric Ice Clouds over Titan's Winter Pole with the Titan Atmospheric Model", 54th Division for Planetary Sciences Meeting

2023. **Lombardo, N. A.** & Lora, J. M. "Titan's Stratospheric Heat and Momentum Budgets", Titan Through Time VI

2022. **Lombardo, N. A.** & Lora, J.M. "Interaction of Chemical Tracers with Titan's General Circulation", American Geophysical Union

2022. **Lombardo, N. A.** & Lora, J. M., "The Energy and Momentum Budgets of Titan's Stratospheric Polar Vortex", 53rd Division for Planetary Sciences Meeting

2022. **Lombardo, N. A.** & Lora, J. M., "The Energy and Momentum Balance of Titan's Stratospheric Polar Vortex as Simulated in a General Circulation Model", Europlanet Science Congress Meeting

2019. **Lombardo, N. A.**, et al., "Detection of Allene on Titan," Europlanet Science Congress – Division for Planetary Sciences Joint Meeting

2019. **Lombardo, N. A.**, et al., "Detection of Allene on Titan," Hydrocarbon Worlds NAI Workshop

2018. **Lombardo, N. A.**, et al., "Using Far-Infrared Observations to Probe Ethane Abundance in Titan's Lower Stratosphere," 50th Division for Planetary Sciences Meeting

2018. **Lombardo, N. A.**, et al. "C3HX Hydrocarbon Abundance in Titan's Atmosphere from *Cassini* Infrared Spectra," 49th Lunar and Planetary Science Conference

2017. **Lombardo, N. A.**, et al. "Abundance Profiles for C3 Hydrocarbons in Titan's Atmosphere," 49th Division for Planetary Sciences Meeting

2015. **Lombardo, N. A.**, et al., "Origin of Photoluminescence in Graphene Oxide", APS NES Fall

Select First-Authored Conference and Workshop Posters

2025. **Lombardo, N. A.**, Nixon, C. A., Koskinen, T. T., "Exploring Titan Dynamic Atmosphere with the Habitable Worlds Observatory", Towards the Habitable Worlds Observatory: Visionary Science and Transformative Technology

2024. **Lombardo, N. A.** & Lora, J. M., "Seasonal-Scale Transport of Heat and Momentum in Titan's Middle Atmosphere", 24th Conference on Atmospheric & Oceanic Fluid Dynamics

2024. **Lombardo, N. A.** & Lora, J. M. "Interaction of Trace Species with Titan's General Circulation", 24th Conference on Atmospheric & Oceanic Fluid Dynamics

2023. **Lombardo, N. A.**, et al., "Measurement of Hydrogen Isocyanide (HNC) in Titan's Lower Stratosphere", Titan Through Time VI

2022. **Lombardo, N. A.**, et al., "Variability and Trends in Beaufort Gyre Near-Inertial Wave Activity inferred from Bottom Pressure Measurements, 2003—2021", Ocean Sciences Meeting

2021. **Lombardo, N. A.** & Lora, J.M., "Dynamical Simulations of Titan's Stratosphere using Observationally Derived Molecular Abundance and Aerosol Opacity," Titan Through Time V

2019. **Lombardo, N. A.**, et al., "Detection of Allene in Titan's Atmosphere," Astrobiology Science Conf.

2018. **Lombardo, N. A.**, et al., "The Search for Allene in Titan's Stratosphere using Mid-IR Spectra from the TEXES Instrument on IRTF," American Geophysical Union, Fall Meeting

2018. **Lombardo, N. A.**, et al., "Comparison of Modeled Alkane Abundances in Titan's Atmosphere using Different Spectral Windows with *Cassini* CIRS spectra," Cassini Science Symposium

2018. **Lombardo, N. A.**, et al., "C3 Hydrocarbon Abundance in Titan's Atmosphere with *Cassini* Infrared Spectra", American Astronomical Society, AAS Meeting #231

2017. **Lombardo, N. A.**, et al., "A Sensitive Search for Allene on Titan", Titan Through Time IV

2016. **Lombardo, N. A.**, et al., "Synthesis and Characterization of $CoFe_{2-x}Ni_xO_4$ Magnetic Nanoparticles", APS NES Spring

2016. **Lombardo, N. A.**, et al., "Controllably Inducing and Modeling Optical Response from Graphene Oxide," APS March Meeting

Additional Group Affiliations

2022–2025	Collaborator	JWST Titan Guaranteed Time Observation Team
2018–present	Collaborator	NASA Astrobio. Inst. Habitability of Hydrocarbon Worlds Group

Public Communication and Community Service

Scientific Consultant for NASA Science Visualization Studio
Invited Public Speaker for geology and astronomy at *Bedford Garden Club* (2025), *Ward Pound Ridge Reservation* (2024), *Pound Ridge Land Conservancy* (2023, 2024), *Trailside Museum Children's Summer Ecology Program* (2023), *Trailside Nature Museum* (2022)
Judge at the *Westchester Junior Science and Humanities* Symposium (2024)
Overnight Near-to-Peer Mentor to group of 5 high school students at the GSFC *Boys Night In* (2019)
Volunteer and Exhibitor representing Cassini CIRS and Lunar Reconnaissance Orbiter (LRO); *International Observe the Moon Night* at GSFC (2018-2019); *Awesome-Con* at Washington, DC (2018-2019); *Celebrating Earth Day with NASA at Union Station* (2017-2019); *Space Night* with the Bowie Baysox minor league baseball team (2018); *US Science and Technology Expo* in Washington, DC (2019); *Science Jamboree* at GSFC (2017-2018); *Outer Space in the President's Place* in the Eisenhower Executive Office Building in Washington, DC (2017)
Convener of the CCSU Physics Open House for local high school students (2015, 2016)
Panelist 22nd Annual Science and Technology Brain Trust, US News STEM Solutions Conference (2014)